

QCE Chemistry Units 3&4

Paper 1

General instruction

- Work in this booklet will not be marked.

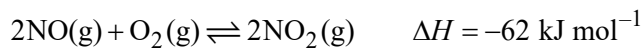
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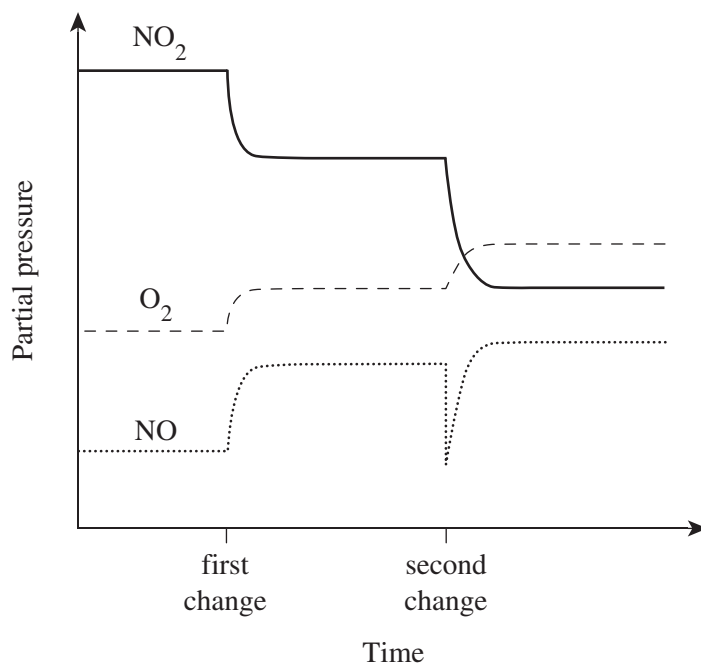
SECTION 1

QUESTION 1

Nitrogen dioxide is formed when nitrogen monoxide undergoes oxidation according to the following equation.



A change was imposed on an equilibrium gas mixture of NO_2 , NO and O_2 . The mixture returned to equilibrium, and another change was imposed. The following graph shows the effects of the two changes.

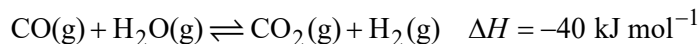


Identify the imposed changes.

	First change	Second change
(A)	temperature decrease	partial pressure of O_2 is increased
(B)	temperature decrease	partial pressure of NO is decreased
(C)	temperature increase	partial pressure of O_2 is increased
(D)	temperature increase	partial pressure of NO is decreased

QUESTION 2

The reaction of carbon monoxide with steam to form carbon dioxide and hydrogen gas is an important industrial process. The reaction occurs according to the following equation.



Which of the following represents the equilibrium constant (K_c) for the reaction?

(A) $K_c = \frac{[\text{CO(g)}][\text{H}_2\text{O(g)}]}{[\text{CO}_2\text{(g)}][\text{H}_2\text{(g)}]}$

(B) $K_c = \frac{[\text{H}_2\text{(g)}]^2 [\text{CO}_2\text{(g)}]}{[\text{CO(g)}][\text{H}_2\text{O(g)}]}$

(C) $K_c = \frac{[\text{CO}_2\text{(g)}][\text{H}_2\text{(g)}]}{[\text{CO(g)}][\text{H}_2\text{O(g)}]}$

(D) $K_c = \frac{[\text{H}_2\text{(g)}][\text{CO}_2\text{(g)}]}{[\text{CO(g)}]}$

QUESTION 3

The table shows the colours of three indicators as the hydrogen ion concentration changes.

$[\text{H}^+] \text{ mol L}^{-1}$	10^{-2}	10^{-4}	10^{-6}	10^{-8}	10^{-10}
phenolphthalein	colourless	colourless	colourless	colourless	pink
methyl red	pink	pink	yellow	yellow	yellow
phenol red	yellow	yellow	red	red	red

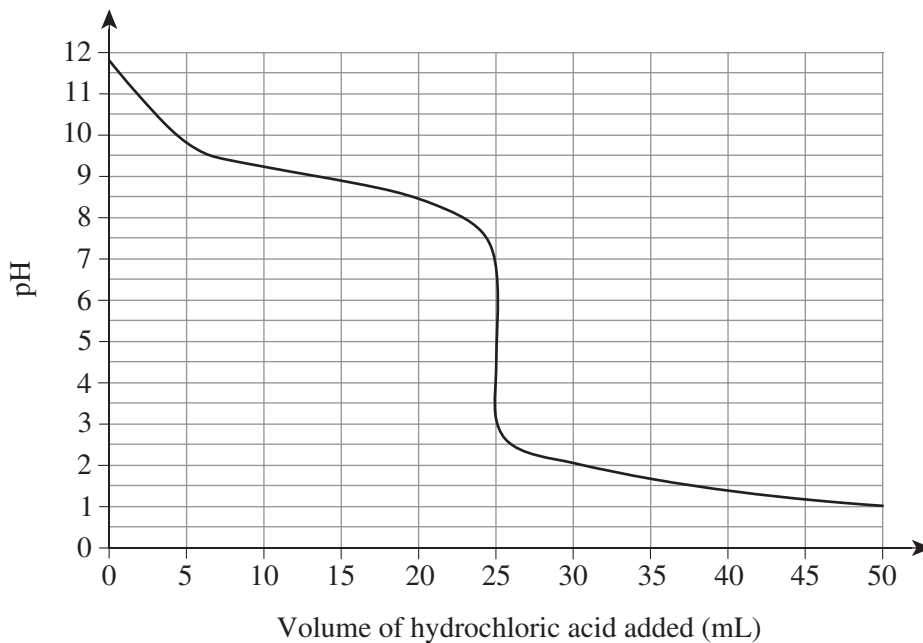
A solution is pink when tested with phenolphthalein, yellow with methyl red and red with phenol red.

Which of the following statements about the solution is correct?

- (A) The solution is basic with a pH of 6.
- (B) The solution is acidic with a pH of 6.
- (C) The solution is basic with a pH of 10.
- (D) The solution is acidic with a pH of 10.

QUESTION 4

The titration curve obtained when 0.1 M hydrochloric acid is added to 0.1 M ammonia is shown.



The equivalence point occurred at

- (A) pH 5.2
- (B) pH 7.0
- (C) pH 9.2
- (D) pH 11.0

QUESTION 5

Which of the following statements regarding the overall three-dimensional structure of proteins is correct?

- (A) Polar neutral amino acids interact through hydrogen bonding.
- (B) Non-polar amino acids interact through hydrophilic interactions.
- (C) Polar acidic and basic amino acids interact through covalent bonds, which is called a salt bridge.
- (D) A disulfide bridge is formed through the bonding of nitrogen atoms in the amino acid cysteine.

QUESTION 6

Determine the oxidation number of S in H_2SO_4 .

- (A) +2
- (B) +4
- (C) +6
- (D) +8

QUESTION 7

Which of the following is the conjugate base of the H_2PO_4^- ion?

- (A) HPO_3^{2-}
- (B) HPO_4^{2-}
- (C) H_3PO_3
- (D) H_3PO_4

QUESTION 8

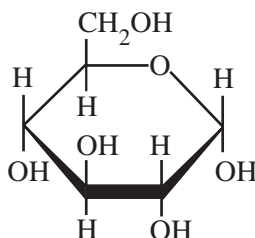
Octan-1-ol and propan-1-ol are both alcohols and are colourless liquids at standard conditions. Octan-1-ol has a boiling point of 195°C and propan-1-ol has a boiling point of 97°C .

Which of the following statements best describes the reason for the difference in boiling points?

- (A) Octan-1-ol has a lower density than propan-1-ol and, therefore, has stronger intermolecular forces.
- (B) Octan-1-ol has a larger hydrocarbon tail than propan-1-ol and, therefore, has stronger intermolecular forces.
- (C) Propan-1-ol has a larger hydrocarbon tail than octan-1-ol and, therefore, has stronger intermolecular forces.
- (D) Propan-1-ol has stronger dispersion forces than octan-1-ol as its neighbouring $-\text{OH}$ groups are close together.

QUESTION 9

The molecule shown is α -D-glucose.



Which of the following statements is correct?

- (A) Glucose is a disaccharide.
- (B) Glucose has three isomers.
- (C) Glucose has the empirical formula CHO .
- (D) Glucose is insoluble in water.

QUESTION 10

Identify the reaction where the equilibrium lies furthest to the right.

	Reaction	K_c
(A)	$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$	1.2×10^{-2}
(B)	$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$	1.6×10^2
(C)	$2\text{H}_2(\text{g}) + \text{S}_2(\text{g}) \rightleftharpoons 2\text{H}_2\text{S}(\text{g})$	1.1×10^7
(D)	$2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{CO}_2(\text{g})$	2×10^{11}

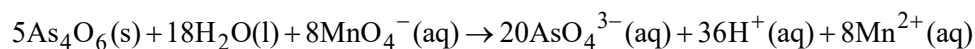
QUESTION 11

Which of the following lists the compounds in order of decreasing boiling point?

- (A) ethane, propanone, propane, propan-1-ol
- (B) propan-1-ol, ethane, propane, propanone
- (C) propanone, propane, ethane, propan-1-ol
- (D) propan-1-ol, propanone, propane, ethane

QUESTION 12

Consider the chemical equation for a reaction.

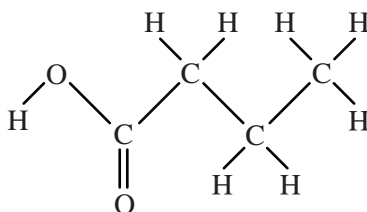


Determine which of the following statements is correct.

- (A) The manganese has been reduced and acts as a reducing agent.
- (B) The arsenic has been oxidised and acts as an oxidising agent.
- (C) The oxygen has undergone oxidation.
- (D) The permanganate ion acts as an oxidising agent.

QUESTION 13

Consider the structural formula of a molecule.



Deduce the name of the molecule.

- (A) butanol
- (B) propanol
- (C) butanoic acid
- (D) ethanoic acid

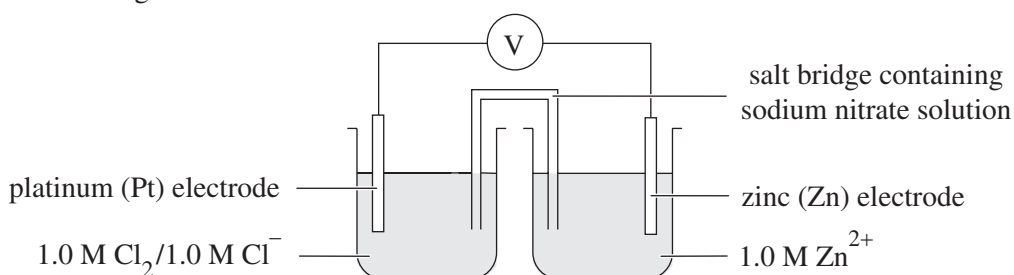
QUESTION 14

Which of the following statements about polymers is correct?

- (A) Addition polymers can only be constructed from saturated monomers.
- (B) To polymerise, condensation monomers must contain only one functional group.
- (C) In a condensation reaction, the units of starch are joined together by glycosidic bonds between the α -glucose monomers.
- (D) Proteins are formed when amino acids undergo addition polymerisation, creating peptide bonds.

QUESTION 15

The diagram shows a galvanic cell.



In the cell,

- (A) reduction occurs at the zinc electrode.
- (B) the platinum electrode will become thinner.
- (C) electrons flow from the zinc electrode to the platinum electrode.
- (D) sodium ions migrate to the half-cell on the right and nitrate ions migrate to the half-cell on the left via the salt bridge.

QUESTION 16

What type of reaction is needed to convert ethanol to ethanoic acid?

- (A) a reduction reaction using phosphoric acid (H_3PO_4) as a catalyst and heat
- (B) an oxidation reaction using H_3PO_4 as a catalyst and heat
- (C) a reduction reaction using acidified dichromate ($\text{Cr}_2\text{O}_7^{2-}$) as a catalyst and heat
- (D) an oxidation reaction using $\text{Cr}_2\text{O}_7^{2-}$ as a catalyst and heat

QUESTION 17

Identify the redox reaction.

- (A) $\text{H}_2\text{CO}_3(\text{aq}) + \text{Ca}(\text{NO}_3)_2(\text{aq}) \rightarrow 2\text{HNO}_3(\text{aq}) + \text{CaCO}_3(\text{s})$
- (B) $2\text{HCl}(\text{aq}) + \text{Mg}(\text{s}) \rightarrow \text{H}_2(\text{g}) + \text{MgCl}_2(\text{aq})$
- (C) $\text{KOH}(\text{aq}) + \text{HCl}(\text{aq}) \rightarrow \text{KCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- (D) $\text{AgNO}_3(\text{aq}) + \text{HCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{HNO}_3(\text{aq})$

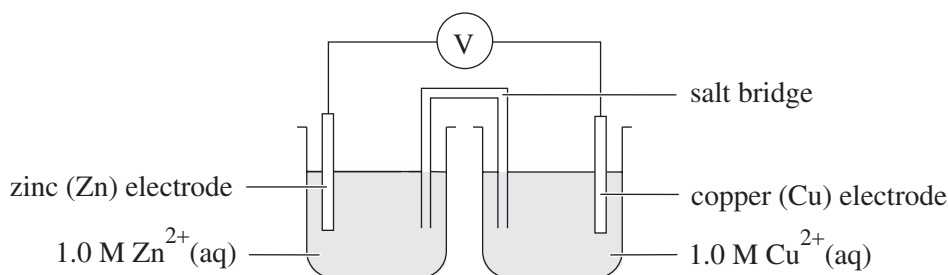
QUESTION 18

Deduce the type of reaction that would change an alkene into an alkane.

- (A) substitution with HCl
- (B) reduction using an Ni catalyst
- (C) oxidation using an MnO_4^- catalyst
- (D) condensation using an Ni catalyst

QUESTION 19

A simplified diagram of a Daniell cell, which is a type of galvanic cell, is shown. The electrolyte in the salt bridge is a potassium nitrate solution, $\text{KNO}_3(\text{aq})$.

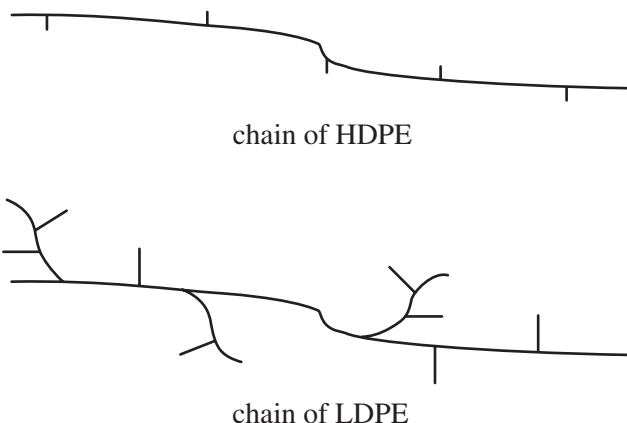


Which of the following statements is correct?

- (A) The zinc electrode is the negative electrode, and the potassium ions flow from left to right in the salt bridge.
- (B) The copper electrode is the negative electrode, and the potassium ions flow from left to right in the salt bridge.
- (C) The zinc electrode is the positive electrode, and the potassium ions flow from right to left in the salt bridge.
- (D) The copper electrode is the positive electrode, and the potassium ions flow from right to left in the salt bridge.

QUESTION 20

The diagrams represent typical polymer chains of the two forms of polyethene: high-density polyethene (HDPE) and low-density polyethene (LDPE).



Which of the following statements is correct?

- (A) In LDPE, the chains are highly branched, resulting in close packing of the polyethene chains.
- (B) LDPE is harder and less flexible than HDPE and can be used to make pipes.
- (C) In HDPE, the chains can pack closely together, forming a rigid, highly crystalline material.
- (D) LDPE is relatively crystalline in comparison to HDPE.

QUESTION 21

The products of the electrolysis of molten sodium chloride and the electrolysis of a dilute aqueous solution of sodium chloride were compared.

Which of the following statements about both cells is correct?

- (A) The cells have the same anode and cathode products.
- (B) The cells have the different anode and cathode products.
- (C) The cells have the same anode product, but different cathode products.
- (D) The cells have the same cathode product, but different anode products.

QUESTION 22

A sodium hydroxide solution (NaOH) was titrated against different acids of 0.100 M concentration until the neutralisation points were reached.

Identify the acid that would require the greatest volume of NaOH to reach the neutralisation point.

- (A) nitric acid (HNO_3)
- (B) carbonic acid (H_2CO_3)
- (C) hydrochloric acid (HCl)
- (D) ethanoic acid (CH_3COOH)

QUESTION 23

A student is testing the reactivity of different metals when the metals are placed in an aqueous solution of copper sulfate. The same sized pieces of each metal are placed in separate beakers and covered with the solution. The table shows the metals and their beaker labels.

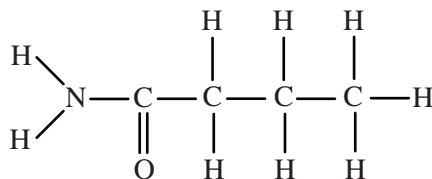
Beaker	Metal
W	Zn
X	Cu
Y	Mg
Z	Mn

Predict which beaker will show no reaction.

- (A) beaker W
- (B) beaker X
- (C) beaker Y
- (D) beaker Z

QUESTION 24

The structural formula of an organic molecule is shown.

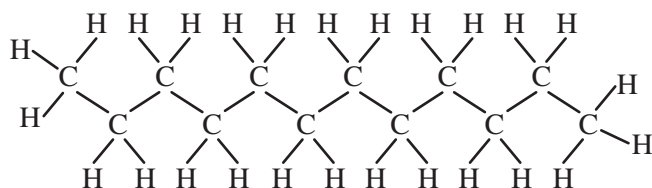


Identify the molecule.

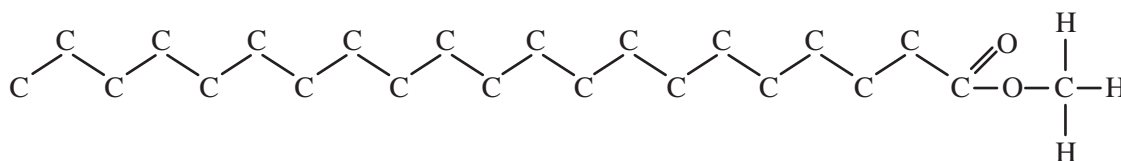
- (A) butanamide
- (B) butanoic acid
- (C) butylamine
- (D) butanol

QUESTION 25

When petrodiesel and biodiesel are compared, their molecules have similar energy contents of 47 kJ g^{-1} . The molecules, however, have different physical properties due to their structures. Examples of a petrodiesel molecule and a biodiesel molecule are shown.



petrodiesel molecule



biodiesel molecule

Which of the following statements is correct?

- (A) Biodiesel has polar bonds, so it has a lesser tendency than petrodiesel to absorb water.
- (B) Biodiesel contains smaller, non-polar molecules than petrodiesel, resulting in a lower viscosity than that of petrodiesel.
- (C) Biodiesel has a higher viscosity than petrodiesel, so it does not flow as easily along fuel lines and through filters.
- (D) Biodiesel releases more energy per litre of fuel than petrodiesel.